Weekly Business Review — Olist Marketplace

VP Executive Memo

78.93%	93.34%	4.16 / 5	3.04%
Satisfied orders	On-time delivery	Avg review score	Repeat purchase

KEY FINDING

Late delivery is strongly associated with lower customer satisfaction. Only 1.86% of 5-star orders were delivered after the estimated date, compared with 36.68% of 1-star orders.

Across 95,824 delivered orders, the marketplace achieved 93.34% on-time delivery and a 4.16/5 average review score.

EVIDENCE

- 1-star reviews: 36.68% late-delivery rate
- 2-star reviews: 18.84%
- 3-star reviews: 8.78%
- 4-star reviews: 3.43%
- 5-star reviews: 1.86%

The observed late-delivery rate for 1-star orders is roughly 20 times the rate observed for 5-star orders.

RECOMMENDATION

Prioritize a delivery-exception management program. Flag orders at high risk of missing their estimated delivery date before the promised date. Route these orders to an operational exception queue for carrier escalation or proactive customer communication.

Track late-delivery rate and 4–5-star satisfaction weekly by carrier, state and delivery lane to identify where intervention has the greatest impact.

The analysis shows association, not proven causation.

North Star: satisfied delivered orders = delivered orders with a 4–5 star review / delivered orders with a review.

Data Quality & Pipeline

DATA QUALITY CHECKS

Check	Failing rows
Null order_id	0
Duplicate order_id	0
Delivery before purchase	0
Orders without customer	0
Payment mismatch	387

Four checks passed with zero failing rows.

PAYMENT MISMATCH HANDLING

The payment reconciliation check identified 387 orders
where payment_total did not equal item_total + freight_total.

These records were FLAGGED rather than dropped or modified. Original values remain available for investigation, while mart_orders contains an explicit payment_mismatch_flag. This preserves completeness and prevents potentially valid payment-related records from being silently removed.

PIPELINE DESIGN

1. Raw CSV ingestion
2. Required-file and required-column validation
3. Cleaned staging tables
4. One-row-per-order analysis mart
5. SQL data-quality validation
6. Pipeline stops with an error when a check fails

The pipeline is idempotent: raw, staging and mart tables use
CREATE OR REPLACE TABLE, so reruns do not append duplicate data.

FAILURE HANDLING

The end-to-end pipeline reached the quality stage and intentionally
stopped because the payment mismatch check detected 387 failing rows.

SQL Analysis & Metric Definitions

PART C — SQL RESULTS

1. Monthly revenue and MoM growth
   Uses LAG() to compare each month with the previous month.
   August 2018 revenue (last complete month analysed):
   ₹848,860.10

2. Top 3 product categories by customer state
   Uses DENSE_RANK() partitioned by customer state.

3. 7-day rolling average of daily orders
   Uses a continuous calendar so zero-order days are included.

4. Repeat-purchase rate
   Uses customer_unique_id across multiple orders.
   Result: 3.04%

5. Delivery performance vs review score
   1-star late rate: 36.68%
   5-star late rate: 1.86%

PART D — METRIC DEFINITIONS

NORTH STAR — SATISFIED DELIVERED ORDER RATE
Formula: delivered orders with a 4–5 star review /
delivered orders with a review × 100
Current value: 78.93%
Why: Directly measures satisfactory delivered experiences.

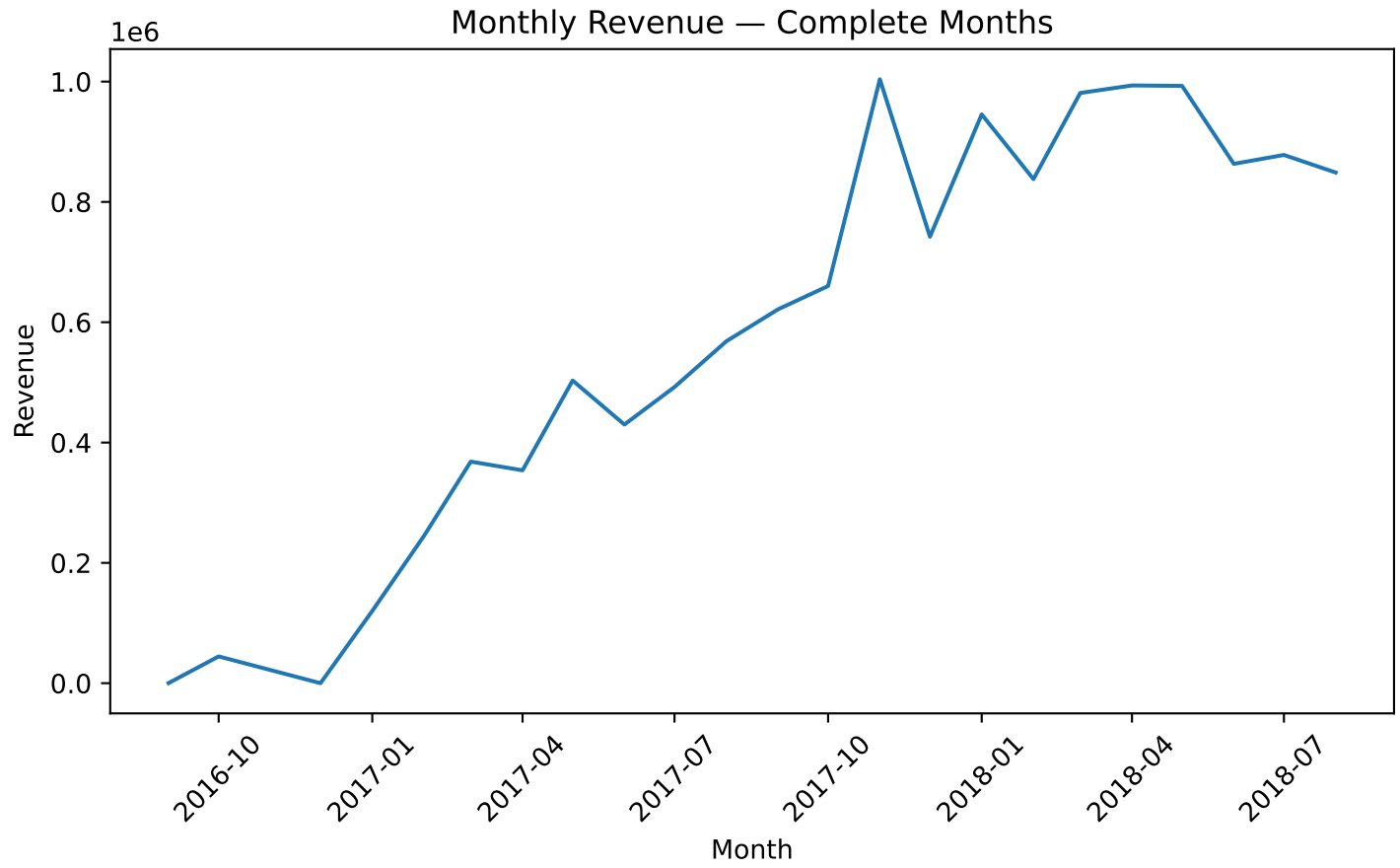
SUPPORTING — ON-TIME DELIVERY RATE
Formula: delivered orders on/before estimated date /
eligible delivered orders × 100
Current value: 93.34%
Why: Measures delivery reliability.

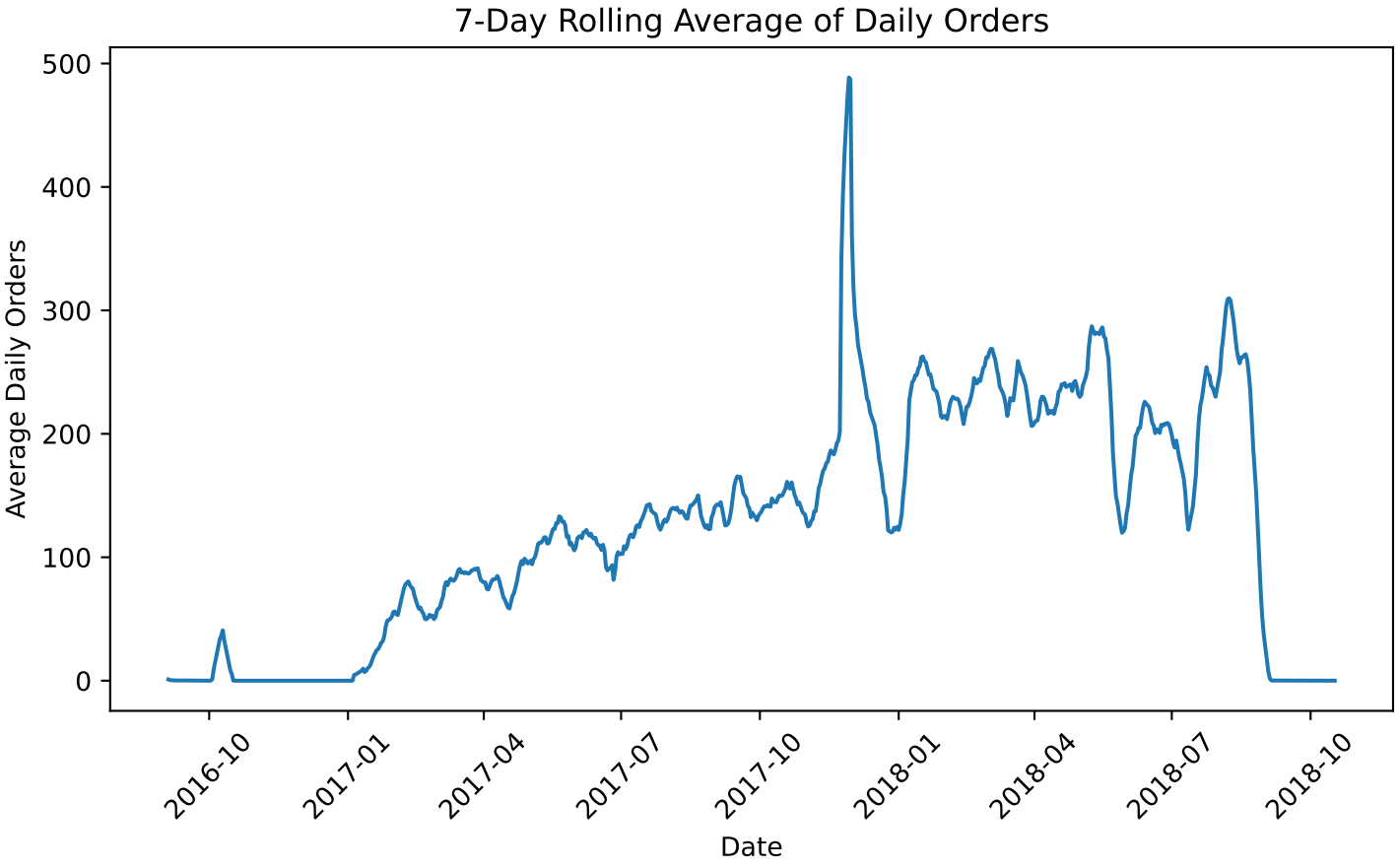
SUPPORTING — AVERAGE REVIEW SCORE
Formula: sum of review scores / reviewed delivered orders
Current value: 4.16 / 5
Why: Measures overall customer experience quality.

SUPPORTING — REPEAT PURCHASE RATE
Formula: repeat customers / qualifying customers × 100
Current value: 3.04%
Why: Measures whether customers return.

GUARDRAIL — MONTHLY REVENUE
Formula: sum of item price across qualifying orders.
Why: Ensures customer-experience improvements are considered alongside commercial performance.

Business Trends





Customer Experience



